PROJECT PROPOSAL

A Visual and Audio Processing System for Real-Time Cello Posture and Performance Feedback

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To Dr. Hashtrudi - Zaad,

On behalf of Lachlan Death, Cohen Little, Maxwell Sloan, and Christian Vandermeer, this team is pleased to submit the enclosed proposal, *A Visual and Audio Processing System for Real-Time Cello Posture and Performance Feedback* in response to the proposal call, *Music Tutor: Posture and Bowing,* for the 2026-2027 MREN 403 Mechatronics & Robotics Design IV course.

This proposal presents a real-time feedback system for beginner cellists practicing independently, structured around three integrated subsystems. The visual processing subsystem, Section 2.4, pairs a pretrained body-and-hand landmark model with a custom-trained detector for the bow, bridge, and fingerboard. Using a hybrid representation bounding boxes shall be used for orienting for the rigid reference objects, key points for the bow, chosen to balance detection precision with available data. The audio processing subsystem Section 2.3, performs real-time pitch and rhythm detection using a YIN algorithm, as shown in Figure 1. Our window-size and latency analysis confirms that pitch can be resolved within the 80ms threshold at which feedback is perceived as instantaneous, with processing margin to spare. The Graphical User Interface, Section 2.5, consumes the output of both pipelines and translates the visual real-time feedback to an augmented overlay on a live mirror feed, a design chosen deliberately to reflect how cellists already practice.

The proposed design choices are grounded in established theory, and methodologies have been selected to deliver effective and intuitive feedback to the user. Every interface decision, from the mirrored feedback, layered overlay and audio feedback cues, prioritizes keeping the player's attention on the betterment of playing rather than on the tool itself. We present this work as a research prototype demonstrating the feasibility of real-time, multimodal feedback for cello technique. The focus is on validating that this kind of feedback can be delivered clearly and intuitively, with commercial deployment considerations left to future work.

Thank you for your time reviewing this proposal, the team would welcome any questions and can provide additional detail on any aspect of the proposed solution.

Sincerely,

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1 Introduction

1.1 Executive Summary

This project proposes a real-time multimodal feedback system for beginner cellists practicing independently. The system shall combine computer vision and processing with audio signal processing to evaluate selected aspects of playing technique, including body and arm posture, bow position and orientation, pitch, and rhythm. Feedback shall be presented through a purpose-built graphical user interface (GUI) which offers the ability for a cellist to receive real time feedback on both posture and pitch or separately the ability to play with sheet music and subsequently review posture, pitch, and rhythm.

1.2 Problem Statement

Proper posture is a fundamental component of cello technique, providing the physical foundation for bowing and intonation. For beginner cellists, proper posture is often difficult to achieve and should be addressed early [1]. Beginner cellists may practice for extended periods without continuous access to an instructor, and therefore a typical approach to practicing posture is to use a mirror as a basic independent teaching tool [2]. Although this method provides the student with a way of independently improving, it lacks quantitative feedback and dimensionality of feedback. The goal of this project is to extend the function of this independent practice in providing quantitative correction not only on posture but also on pitch and rhythm. The problem at hand is therefore to provide a system which can provide feedback on cello-playing posture, bowing, pitch, and rhythmic variables with sufficient accuracy and low latency to provide useful real-time feedback during independent beginner practice.

2 Proposed Solution

2.1 Solution Overview

The solution shall have two fundamental modes of operation: open practice and guided practice. Under open practice mode, the user shall play freely and receive overlaid visual feedback on their posture and separate visual feedback on their pitch. In guided practice mode, the user shall play in accordance with displayed sheet music and receive real-time audio feedback overlaid on the sheet music and subsequent to playing, a video containing overlaid feedback for both sound and posture.

2.2 Project Scope

The proposed solution can be most clearly conceptualized as three distinct subsystems: Visual Processing, Audio Processing, and a GUI. Visual Processing and Audio Processing shall operate as independent, parallel processes, both simultaneously capturing the same live cellist performance, with respect to body and bow posture, and pitch and intonation respectively.

2.3 Audio Processing

The Audio Processing subsystem shall provide pitch and rhythm detection. In guided practice mode, these quantities shall allow for played music to be compared to a desired piece of sheet music. This shall occur in real time, allowing the player to see how they are performing as they play. In open practice mode, pitch shall be compared with the nearest note expected on a standard tuned cello. The system shall use a YIN algorithm to detect pitch, and a global clock to detect when notes are played when necessary.

In order to achieve the preceding functional overview, the audio processing system must be capable of capturing audio, processing it, and providing information on pitch and temporal alignment to then be shown on the GUI. The high-level architecture of the audio processing system can be found in Figure 1.

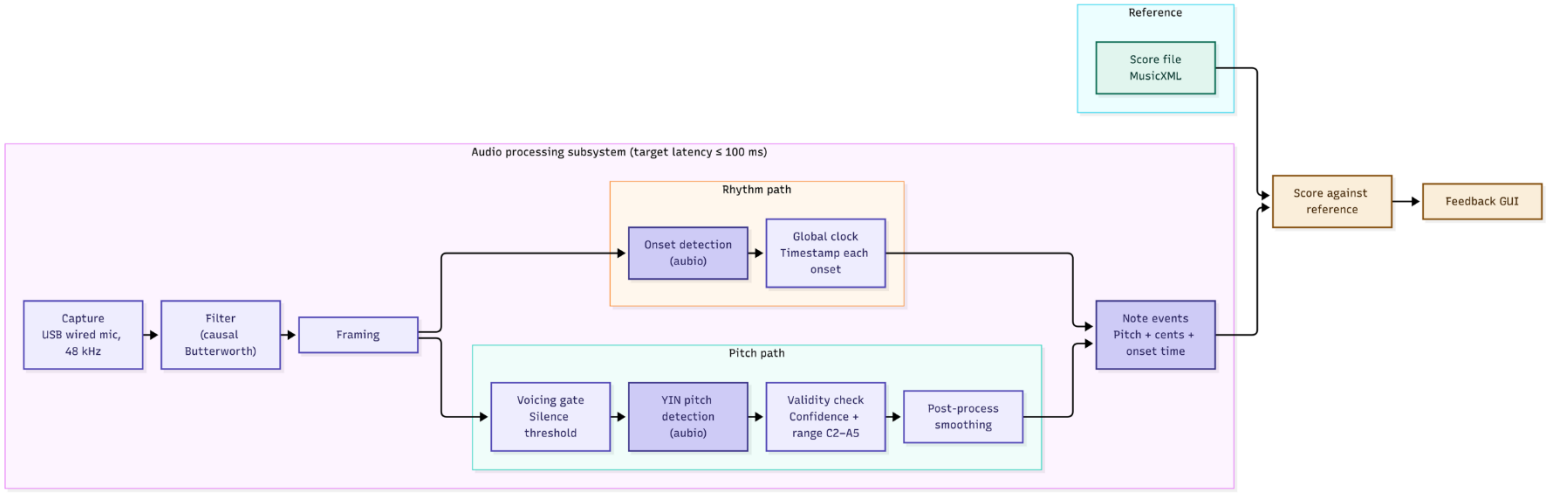


Figure 1. Flowchart of audio subsystem.

2.4 Visual Processing

The Visual Processing subsystem provides real-time feedback on the cellists form. The player shall sit in front of a camera and monitor while playing, and the system shall track the position of their arms, hands, and bow and compare it against correct cello postures. This shall occur in real time, allowing the player to see their form as they play. The resulting posture classifications shall be displayed to the player as a colour-coded overlay on a live, mirrored video feed, with feedback on all relevant postures (bow, arm, instrument, body, etc.). The system will combine two perception models, MediaPipe and YOLOv8 that serve complementary roles. MediaPipe Pose and Hands [3], [4] supply 33 body landmarks and 21 per hand. A custom-trained YOLOv8 detector supplies bow (frog, tip), bridge and fingerboard detections.

To achieve the preceding functional overview, the visual processing system must be capable of capturing video, estimating the posture of the player, classifying that against correct form, and providing the resulting classes to the GUI. The high-level architecture of the visual processing system can be found in Figure 2.

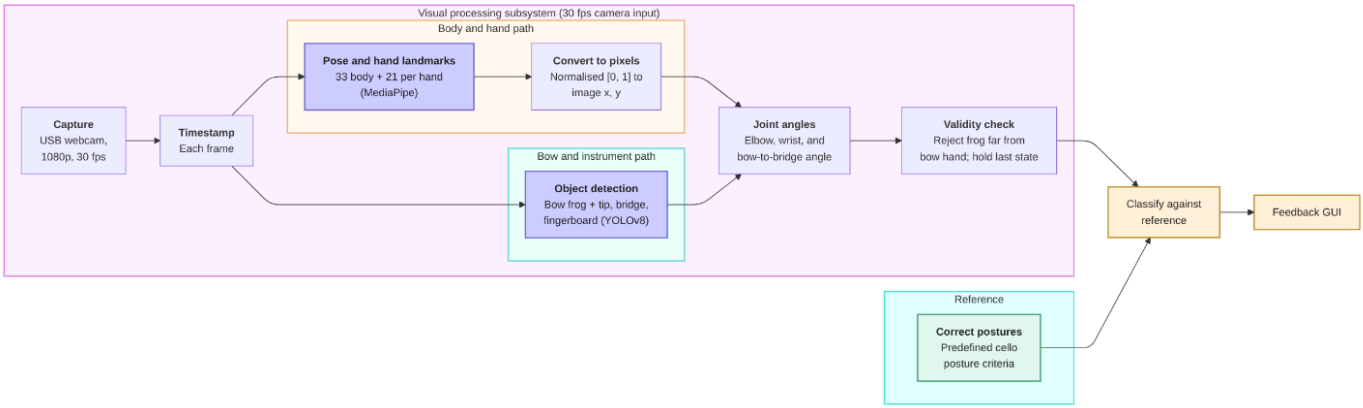


Figure 2. High-level architecture of the visual processing system

2.5 Graphical User Interface

The GUI shall be the point of integration, consuming the classified output of both subsystems and rendering a cognitively ergonomic representation of the user's playing. The GUI shall be capable of providing real time video of a student playing with overlaid graphics relating to their posture, displaying the pitch of a detected note, and displaying pitch and temporal alignment of audio in contrast to sheet music.

3 Review of State of the Art

This section reviews research most relevant to real-time posture assessment and multimodal feedback for bowed-string instruments.

The Purdue Cello Evaluator [5] combines YOLOv11 oriented bounding boxes (OBB) with MediaPipe Pose and Hands to assess cello-specific bowing and upper-body posture. Two small neural networks classify wrist and elbow posture, achieving over 95% validation accuracy and approximately 50 ms inference on a Samsung S25 with neural-processing acceleration. However, bow detection achieved only 78% correct classification, and the system does not assess audio performance. Usability testing also identified feedback transparency and error recovery as areas for improvement.

Blanco et al. [6] evaluated a technologically enhanced music learning project (TELMI) using SkyNote, a novel tool used to offer real-time feedback to violin players. The study evaluated 50 beginner violinists and 15 experts. Microphone audio was processed using YIN to estimate pitch and instantaneous power, with pitch and dynamic stability displayed visually. Separately, a Microsoft Kinect and reflective markers tracked bow movement and orientation. During kinematic feedback, beginners improved bow skewness by 31% relative to baseline. At transfer, the feedback group demonstrated 10.3% greater improvement in dynamic stability than the control group. However, 55.2% of feedback-group participants reported difficulty playing while watching the display, highlighting potential attentional demands. However, this evaluation had no resources for specific GUI or user experience parameters.

Existing work demonstrates cello-specific computer vision and multimodal musical feedback, but with differing functionality and intention. This project integrates methods of both practices, and playing evaluation into one application, focusing on designing for user experience

4 Technical Background

4.1 Video Processing

4.1.1 Design Objective and Assumptions

The video pipeline specified requirements come from the task, the algorithms are chosen to meet them, and the result is checked against a shared latency budget.

For Video processing one camera shall feed two models in parallel. MediaPipe shall supply body and hand landmarks, and a custom YOLOv8 detector shall supply the bow and instrument. A fusion stage shall turn both outputs into joint and bow position and angular data, and the posture shall be classified.

The video subsystem is designed under the following assumptions:

1. One cellist is in frame entirely visible.
2. The camera stays fixed during a session. The Purdue evaluator (and presumably the majority) were also trained from one similar camera position, which limits accuracy from other angles (Wang et al., 2026).
3. Lighting is reasonably even and very consistent, with no strong backlight.
4. Wrist depth is not used. A single RGB camera gives reliable x and y, and Purdue excluded wrist depth because it was unreliable during movement (Wang et al., 2026).
5. The player is a seated beginner playing in first position.

4.1.2 Camera Selection (Hardware)

The Kensington W1050 meets the requirements in Table 1, which are derived the same way as the microphone requirements in Table 2 (Section 4.2).

Table 1: Camera desired metrics

Metric	Requirement	Reason
Resolution	≥ 1920 × 1080	Bow is ~7 px wide; ~2 px downscaled.
Frame rate	≥ 30 fps	33.3 ms per frame; not the bottleneck.
Field of view	Upper body, cello and full bow stroke in frame at 1.5 to 2.0 m	87° horizontal: 2.85 × 1.61 m at 1.5 m.
Focus and exposure	Fixed focus sharp at 1.5 to 2.0 m; exposure lockable	Auto exposure blurs the bow; lock untested.
Interface	Wired USB-A	No wireless delay, as with the microphone.

The W1050 is a 1080p, 30 fps webcam with fixed focus and a 95° diagonal field of view (Kensington W1050 listing, PB Tech). For an image of W × H pixels, an object of size L at distance d spans n pixels.

$$\tan\frac{\theta_h}{2} = \tan\frac{\theta_d}{2}\cdot\frac{W}{\sqrt{W^2+H^2}} \qquad n = \frac{LW}{2dtan(\frac{\theta_h}{2})}$$

Equations 1: Camera related equations

4.1.3 Pose, Hand and Object Detection

MediaPipe provides landmarks for the body parts: 33 for the body (BlazePose detects, then tracks) and 21 per hand (Bazarevsky et al., 2020; Zhang et al., 2020), Purdue used it for cello posture (Wang et al., 2026). Limits: the head must be visible, and the cello occludes the body. It cannot see the bow, bridge or fingerboard, so something else is trained for them.

To detect non bodily key points Purdue compared (SAM 3, Detectron2, RT-DETR, YOLO), YOLO was most effective and thus selected (Wang et al., 2026). YOLO detects in one pass (Redmon et al., 2016) and is scored by IoU, AP and mAP:

$$IoU = \frac{|B_p \cap B_g|}{|B_p \cup B_g|} \qquad AP = \int_0^1 p(r)dr \qquad mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i$$

Equations 2: Computer vis/Video processing related equations

mAP50 counts a detection as correct at IoU ≥ 0.5 (Ultralytics documentation). Purdue found keypoints inaccurate at their dataset size and instead used OBB (oriented bounding box) (Wang et al., 2026). This project keeps the one with the lower error and may use YOLOv11.

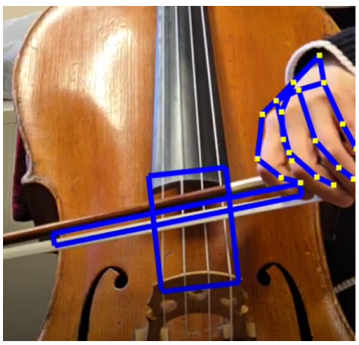


Figure 3. Implemented YOLO methods for determining bow height and angle relative to strings [5]

4.1.4 Angles processing and validity

All geometry is computed in pixels. MediaPipe returns x and y normalized to 0, 1, and at 1920 × 1080 a 45° limb would appear at roughly 61° if left normalized.

- The joint angle θ at B uses three landmarks A, B and C, such as shoulder, elbow and wrist.

- The bow vector b runs from frog f to tip t, s is the bridge direction, and δ is the bow's deviation from parallel to the bridge.

$$\theta = \arccos\left(\frac{(A-B)\cdot(C-B)}{\|A-B\|\|C-B\|}\right) \qquad \alpha = \operatorname{atan2}(s \times b, s \cdot b) \qquad \delta = \min(|\alpha|, \pi - |\alpha|)$$

Equations 3: Angles/Vectors related equations

4.1.5 Classification

The angles, unit direction vectors and δ feed a classifier. Purdue used networks of about 440 and 1,100 parameters (Wang et al., 2026). Criteria such as elbow height and wrist position come from literature (Orme-Stone, 2020) and an instructor. Accuracy shall be reported as a confusion matrix on held-out sessions, split by session not frame.

4.2 Audio Processing

4.2.1 Design Objective and Assumptions

The audio system will estimate pitch and temporal alignment from the cellist's performance and provide feedback through the GUI. A YIN pitch detection algorithm shall be used, selected for its computational efficiency and accuracy in estimating the fundamental frequency of monophonic audio. The subsystem is designed to meet an end-to-end feedback latency of fewer than 100 ms while operating concurrently with the video processing system on the same computer. The specifications of the system can be found in Section 5.1.

The audio subsystem is designed under the following assumptions:

1. Monophonic inputs: one cellist plays at a time, monophonically (no double stops or chords).
2. Standard tuning: the cello is tuned to standard pitch, with A4 at 440 Hz [ISO 16].
3. Controlled acoustic environment: playing takes place in a reasonably quiet room, without other instruments or loud speech.
4. Fixed microphone position: the microphone remains in a fixed position relative to the cello during a session.

4.2.2 Hardware Requirements

A 48kHz USB wired microphone shall be used to capture audio. A wired connection is preferred and better suited for this application of real-time audio processing for its lack of delay compared to wireless options.

Microphone specifications: A microphone is considered to be viable for the audio processing system if it meets the metrics in Table .

Table 2: Microphone desired metrics

Metric	Requirement	Reason
Low-frequency response	Usable to ≤ 50 Hz (ideally within −3 dB at 60Hz)	The lowest note, C2, is 65.4 Hz.
High-frequency response	Flat to ≥ 5 kHz	Covers fundamentals up to A5 (880 Hz).
Sample rate	Supports 48 kHz	Neatly fits 2048 samples into 2.8 periods of C2.
Bit depth	≥ 16-bit	Enough dynamic range for soft playing.
Connection interface	Wired: USB	No drivers needed on Ubuntu or Windows.
Input latency	≤ 10ms (mic + driver buffer)	Keeps latency small.
Maximum SPL	≥ ~110 dB SPL before clipping.	A cello played loudly can roughly reach 100–110 dB SPL [7].

In alignment with Table , the proposed hardware system consists of an Audio-Technica PRO 35 clip on microphone connected to a Focusrite Scarlett Solo USB audio interface.

4.2.3 Audio Processing Pipeline

The audio subsystem is required to detect pitches from C2 to A5 (65.41 Hz to 880 Hz) [8]. Preprocessing shall be implemented with a band-pass filtering stage using a Butterworth filter. A separate amplitude-based gate will be used before passing audio into subsequent layers of the pipeline, blocking unwanted low amplitude readings which do not represent the anticipated cello audio.

Subsequent to the amplitude gate, the audio shall be windowed. Pitch estimation requires observing a waveform over a sufficient number of samples to determine its fundamental frequency. For the purpose of this system, this number of samples must contain two to three periods of any desired fundamental frequency [9]. The sampling rate is therefore bound by the lowest frequency pitch desired to detect, which for the cello is C2 at 65.41 Hz. At 48 kHz, a 2048 sample window has a duration of approximately 42.7 ms. The period of C2 is approximately 15.3 ms, allowing for a sampling rate of 48 kHz to sample 2.8 periods of C2. Therefore, a sampling rate of 48 kHz is sufficient to estimate any expected frequency played by a cello and was chosen to fit comfortably within the range of samples sufficient for pitch detection.

The YIN algorithm shall be used to estimate the fundamental frequency of each analysis window. YIN is an algorithm based on autocorrelation that estimates the frequency of a note by comparing an analysis window with delayed copies of itself [10]. The algorithm was selected for its low computational cost, and suitability for monophonic audio. Although neural-network models such as CREPE are available, suitable and sufficient data for their use is not available or worthwhile obtaining [11].

Each pitch estimate will undergo two validation checks before being used for feedback. First, YIN shall provide a confidence value based on how clearly the waveform repeats. Estimates below a set confidence of 0.8 and pitches outside of the cellos fundamental frequency range shall be rejected. Each estimate shall subsequently be rounded to the nearest note. Every hop, the system shall consider the most recent 5 estimates and display the most frequent note. An odd number of estimates shall be used to avoid ties. The exact pitch values for the output note shall be used to calculate the pitch deviation, in cents.

For temporal analysis onset detection shall be used to find the beginning of each note which is different from the previous note. The system shall use a spectral flux onset detector shall be used. Spectral flux detects increases in spectral magnitude between successive frames, allowing for changes in spectral content to be detected.

4.3 Graphical User Interface

A primary feature of the GUI is a mirrored representation of the player. This design mimics common practice of beginner cellists, which is playing in front of a mirror. This technique seeks to improve features of playing that do not necessarily correlate with noticeable change in the produced sound quality but rather reduce playing effort and improved control. [2] The second key feature is augmented segments overlaid on the mirrored display, that follow the orientation and position of the players upper arm, forearm and hand. These segment shapes shall be capsule-continuous, to guide the players focus, following Gestalt principles. [12] Colour-coded feedback shall also be used on these segmentations (green = acceptable posture, red = poor posture). This approach of familiar cuing reduces cognitive load ensuring beginner cellists can maintain focus on indicated corrections rather than interpretation of the interface. [13] This same approach shall be used in the pitch and rhythm detection, to augment the quality of the playing over selected sheet music.

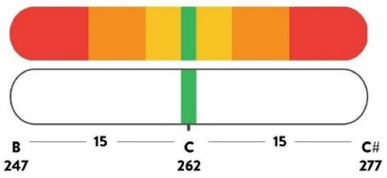


Figure 4. Example pitch meter using Gestalt UI Principles [13]

5 Project Management and Evaluation

5.1 Audio

The following metrics outline the desired metrics within the audio processing system.

Latency: 80-120ms is a standard tolerable audio latency limit, as used by third person video games [14]. The desired maximum latency is therefore 100ms.

Intonation: The system shall measure the pitch of sustained notes to within 2 cents, based on the 1 cent accuracy of a standard chromatic tuner such as the Korg CA-1 [15], with additional tolerance for real time processing.

Onset detection: The system shall detect at least 83% of note starts within 50ms, with no more than 4.1% false detections, matching the published performance of spectral flux onset detection which shall be used [16].

Correctness criteria: A note shall be marked as correct if it is the expected note, its median pitch over the steady part of the note is 4 cents of the target, and its onset is within the smaller of 100ms. These tolerances are twice as wide as the system's measurement accuracy (2 cents and 50ms) so that correctly played notes are not marked wrong due to measurement error but still maintain high accuracy.

5.2 Video

The following metrics outline the desired metrics within the video processing system.

Latency: Like audio, and for the same reasons, the maximum latency for this project is 100ms.

Landmark reliability: The MediaPipe baseline shall run on the webcam or camera feed at 30 fps or better, with the landmarks above the visibility threshold in at least 90% of frames.

Object detection: The YOLOv8 model shall reach an mAP50 of at least 0.80 on the bridge and fingerboard boxes. It shall detect the bow in at least 80% of test frames, against the 78% reported by the Purdue Cello Evaluator (Wang et al., 2026). It shall also localize the bow frog and tip to within 3% of the image width.

Bow angle: The bow angle relative to the bridge axis shall have a mean absolute error of 5° or less. This will be tested on >100 test frames.

Posture classification: The classifier shall label frames with at least 85% accuracy on held-out sessions. It will be tested on a labelled test set that excludes the recording sessions used for training.

6 Standards, safety and ethics

For carrying out usability testing with cellists, this project requires Queen’s General Research Ethics Board (GREB) clearance, following TCPS 2. Additionally, any video of people is personal information, so state that video is processed locally and not stored or transmitted without consent (PIPEDA). The only physical hazard imposed is USB-powered devices, cable routing and trip hazards during testing.

7 Schedule of Work

Discipline	Deliverable	Due
Visual	Evaluations and selection between YOLOv8 and YOLOv11	Nov 1, 2026
Team	Review and submission of project portfolio I	Nov 9-13, 2026
Team	Review and submission of preliminary design (PDR)	Nov 30-Dec 4, 2026
Visual	First model for classifications for posture	Dec 4, 2026
Audio	YIN pitch detection with validity check, majority-vote smoothing and cents offset	Jan 15, 2027
Team	Review and submission of project portfolio II	Jan 25-29, 2027
Audio	Onset detection (spectral flux plus pitch-change) and note scoring against the sheet music	Jan 29, 2027
Visual	Posture correction, its training data, posture correction enforced with thresholds and sources	Feb 1, 2027
GUI	Feedback rules and settings (threshold, fade, off)	Mar 1, 2027
Team	Critical Design Review (CDR)	Mar 1-5, 2027
Audio	Audio evaluation report: latency, note identification, intonation, onset detection	Mar 5, 2027
Team	Prototype Demonstration & Technical Defense	Mar 22-26, 2027
GUI	Complete evaluation of usability	Apr 1, 2027
System	Final report, demonstration, and user guide covering setup and calibration	Apr 23, 2027

8 Budget and Resource

The budget is constrained $500.00 CAD, as per the call for proposals. Current committed expenditures total $ 278.99 CAD, presented in Table . A reserve of $50.00 CAD (10% of the constraint) is held for price fluctuations and hardware replacements, which leaves $171.01 CAD unallocated. All MediaPipe, YOLOv8, and, YIN resources are free and open source for this project's use.

Table 3. Project Bill of Materials (BOM)

Item	Description / Justification	Unit Amount (CAD)
Kensington W1050 Webcam 2 Megapixel 30 fps	1080p @ 30fps webcam. Sufficient for YOLOv8/MediaPipe input, and 95° field-of-view.	69.99
Microphone	Audio-Technica PRO 35.	209.00
USB interface	Focusrite Scarlett Solo USB audio interface	(Owned)

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